

## AI Use Declaration Form

**Course:** Introduction to Artificial Intelligence

**Lab Title:** lab-4-llm-decision-support

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**GitHub Repository:** <https://github.com/Meh02ajv/lab-4-llm-decision-support.git>

**Date Submitted:** Sunday, August 16, 2026

### 1. AI Use Summary

| Question                                            | Student Response |
|-----------------------------------------------------|------------------|
| Did you use any AI tool for this lab?               | Yes              |
| Estimated percentage of the work influenced by AI   | 15%              |
| Did you attach evidence of AI use where applicable? | Yes              |

### 2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

| Name of AI Tool Used | Purpose for Using the Tool                                                       | Prompt or Instruction Given to the Tool                                                                         | Part(s) of the Work Influenced by the Tool | How I Verified, Edited, or Improved the AI Output                                |
|----------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------|
| Gemini AI            | Correction of Errors,<br>Explanation of concepts,<br>Showing the use of syntaxes | <ul style="list-style-type: none"><li>How to go about this question?</li><li>How to fix this problem?</li></ul> | Part 4.1 and Part 3.2                      | By referring to the lab instructions and similar implementations on the internet |
|                      |                                                                                  |                                                                                                                 |                                            |                                                                                  |
|                      |                                                                                  |                                                                                                                 |                                            |                                                                                  |

### 3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

| Evidence Type      | Attached?<br>(Yes, No, N/A) | File Name |
|--------------------|-----------------------------|-----------|
| AI-generated draft | No                          |           |
| Prompt history     | No                          |           |

|                                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Screen<br>shot of AI<br>interaction | Yes | <div><div><pre>from google.colab import userdata GROQ_API_KEY = userdata.get('GROQ_API_KEY')</pre></div><p>The <code>KeyError: 'GROQ_API_KEY'</code> occurs because Python cannot find an environment variable named <code>GROQ_API_KEY</code> in your system environment or inside a loaded <code>.env</code> file.</p><p><b>Option 1: Create or update your <code>.env</code> file (Local Setup)</b></p><ul style="list-style-type: none"><li>Ensure a file named <code>.env</code> exists in the exact same directory as your Jupyter notebook.</li><li>Add your API key inside the file without quotes:</li></ul><div><div>Extrait de code</div><pre>GROQ_API_KEY=gsk_your_actual_api_key_here</pre></div><ul style="list-style-type: none"><li>Re-run <code>load_dotenv()</code> after creating or updating the file.</li></ul><p><b>Option 2: Use Google Colab Secrets (Colab Setup)</b></p><ul style="list-style-type: none"><li>If you are running this in Google Colab, click the <b>Secrets</b> (🔑) icon on the left sidebar.</li></ul></div> <div><div><p>*1. What is the difference between the 'system' and 'user' roles? Give an example of something that belongs in each.*</p><p>*2. What is a token, roughly? Why do API providers bill per token rather than per request?*</p></div><div><div>📄</div><div>✎</div></div><p><b>1. System Role vs. User Role</b></p><ul style="list-style-type: none"><li><b>System Role:</b> Sets the persistent rules, background persona, constraints, and formatting guidelines that govern how the model behaves during the interaction.<ul style="list-style-type: none"><li><b>Example:</b><br/>"You are a microfinance loan assistant. Extract financial data into a valid JSON object without conversational commentary."</li></ul></li><li><b>User Role:</b> Represents the dynamic query, text, or prompt provided at runtime for the model to process.<ul style="list-style-type: none"><li><b>Example:</b><br/>"Extract the loan details from this letter: 'My name is Kofi and I need GHS 3,000 to buy inventory for my provision shop.'"</li></ul></li></ul><p><b>2. Tokens and Per-Token Billing</b></p><ul style="list-style-type: none"><li><b>What is a token?</b> A token is a basic chunk of text—such as a word, sub-word, or punctuation mark—that an LLM uses to process and generate language. As a rule of</li></ul></div> |
| Exported AI conversation            | No  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| AI-generated code snippet           | Yes | <pre>def extract_fields(letter_text):      user_prompt = f"Extract information from this loan application letter:\n\n{letter_text}"      raw_response, _ = ask_llm(         user_prompt=user_prompt,         system_prompt=EXTRACT_PROMPT,</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | <pre> temperature=0.0 )  # Strip markdown code fences if present cleaned_response = raw_response.strip()  cleaned_response = raw_response.strip().removeprefix("```json").removeprefix("```").removesuffix("```").strip()  # Parse JSON and handle failures try:     data = json.loads(cleaned_response)     return data except json.JSONDecodeError as e:     print(f"Warning: Failed to parse JSON response. Error: {e}")     print(f"Raw response received:\n{raw_response}")     return None  for field in fields:     row = {"Field": field}     correct_count = 0      for key in gold_keys:         ext_val = df_indexed.loc[key, field]         gold_val = GOLD[key][field]          # Simple comparison: lowercase strings, exact match for numbers/booleans/None         if isinstance(gold_val, str):             is_match = str(ext_val).strip().lower() == str(gold_val).strip().lower()         else:             is_match = ext_val == gold_val          row[key] = "✓" if is_match else "X"         if is_match:             correct_count += 1      # Calculate accuracy percentage     row["Accuracy"] = f"{correct_count / len(gold_keys):.0%}"     rows.append(row) </pre> |
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|                                       |    |  |
|---------------------------------------|----|--|
| Revised version showing student input | No |  |
|---------------------------------------|----|--|

#### 4. Student Declaration

I declare that:

| Declaration Statement                                                                     | Response (Yes/No) |
|-------------------------------------------------------------------------------------------|-------------------|
| The submitted work is my own work.                                                        | Yes               |
| Any use of AI tools has been clearly declared in this form.                               | Yes               |
| The prompts, instructions, and AI-generated outputs have been disclosed where applicable. | Yes               |
| I reviewed, tested, edited, and improved any AI-generated content before submission.      | Yes               |
| My AI usage does not exceed 25% of the entire work.                                       | No                |
| I understand that undeclared or excessive AI use may be treated as academic misconduct.   | Yes               |

**Student Signature:** Hariel

**Date:** Sunday, August 16, 2026